Firm Entry and AI: Evidence from US Industries

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April 15, 2026

Preliminary and incomplete. Comments welcome.

Abstract

In this short paper I show that, following the release of ChatGPT in November 2022, firm entry increased in AI-exposed sectors, with effects becoming persistently statistically significant from May 2024 — approximately 18 months after the ChatGPT launch. Using monthly sector-level business application data from the Census Bureau Business Formation Statistics and a difference-in-differences design, I find that the effect is [X]% larger for a sector at the 75th versus 25th percentile of AI exposure. The results suggest that AI lowers barriers to entry rather than entrenching incumbents.

Keywords: business formation, artificial intelligence, AI exposure, difference-in-differences, ChatGPT

JEL codes: L26, O33, J24, M13

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1 Introduction

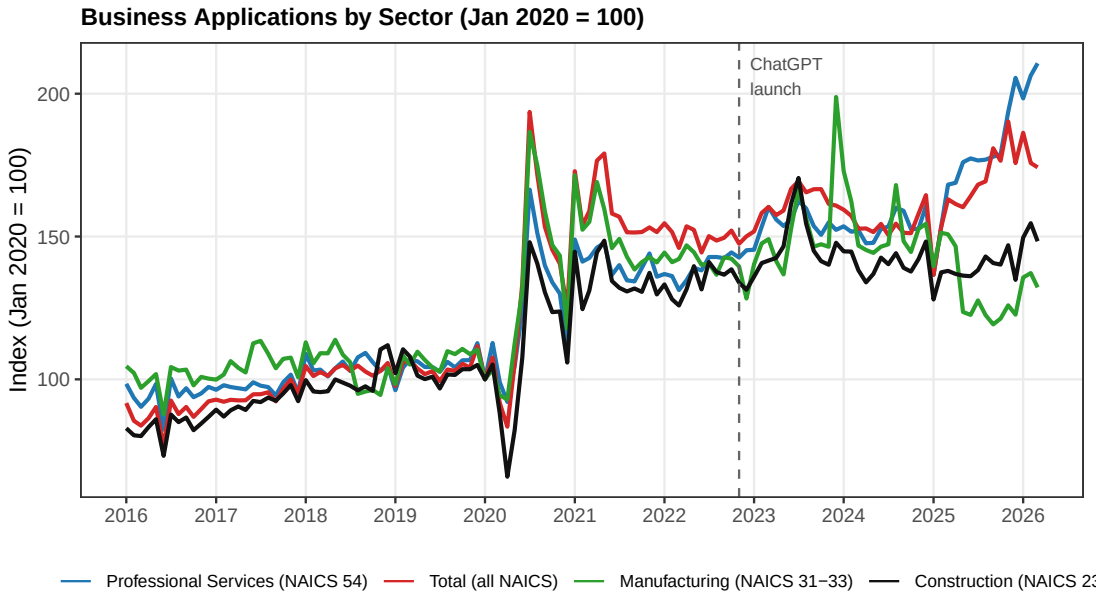

Figure 1: Business applications indexed to January 2020 = 100. Professional Services (NAICS 54), Total (all NAICS), Manufacturing (NAICS 31–33), and Construction (NAICS 23). Dashed vertical line marks November 2022 (ChatGPT launch). Seasonally adjusted. Source: Census Bureau BFS.

After the COVID-19 shock, aggregate business applications surged to historic highs and have continued rising through 2025, but the expansion is not uniform across sectors (Figure 1). Sectors with higher AI exposure have pulled sharply away from the aggregate trend, while goods-producing sectors have lagged—a pattern visible in the raw data and confirmed by the cross-sectional scatter in Figure 2 (as well as the regression in Table X).

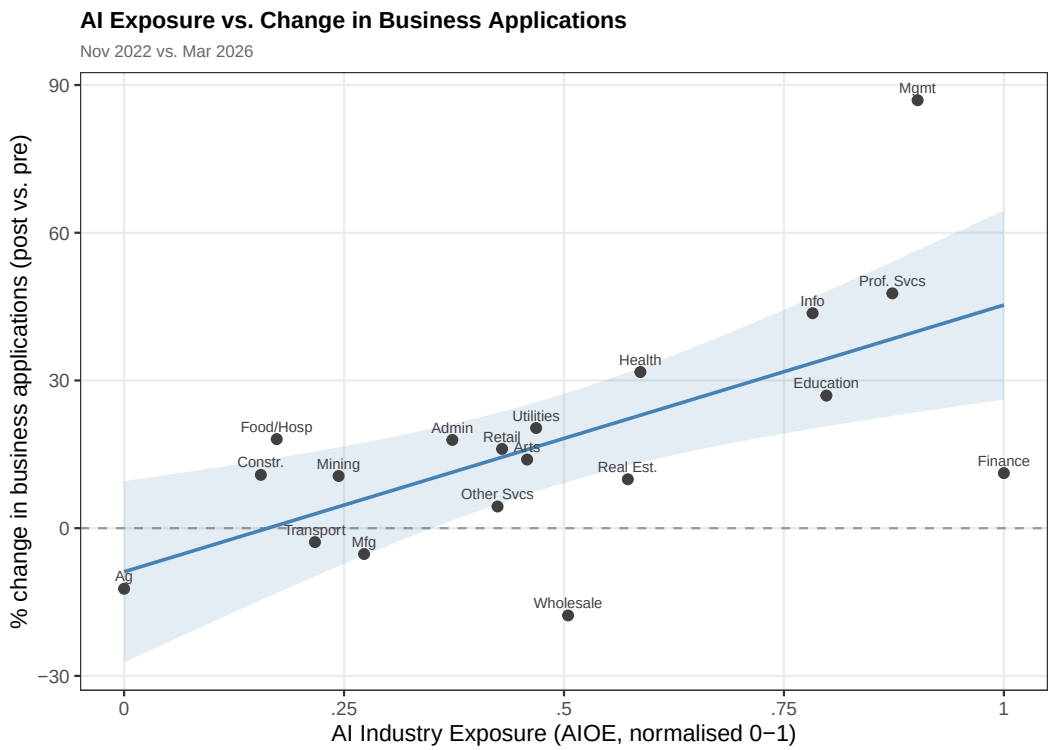

Figure 2: AI exposure versus change in business applications by sector. Horizontal axis: normalised AI Industry Exposure (AIOE). Vertical axis: percentage change from November 2022 to the most recent month available. Each dot is a 2-digit NAICS sector.

We use monthly sector-level business application counts from the Census Bureau Business Formation Statistics (Haltiwanger et al., 2020) together with the AI Industry Exposure (AIIE) index of Felten et al. (2021) to estimate a difference-in-differences (DiD) model of the form

$$\log(\mathrm{BA}_{it}) \;=\; \alpha_i + \gamma_t + \beta\,(\mathrm{Post}_t \times \text{AI Exposure}_i) + \varepsilon_{it}, \tag{1}$$

where i indexes 2-digit NAICS sectors, t indexes calendar months, Post_t is an indicator equal to one from November 2022 onwards, α_i are sector fixed effects absorbing time-invariant differences in application rates, γ_t are month fixed effects absorbing aggregate cyclical variation, and standard errors are clustered by sector. The coefficient β identifies whether sectors with higher AI exposure experienced a differential change in business formation after ChatGPT.

Related literature. The arrival of large language models—most visibly the public release of ChatGPT in November 2022—has revived interest in how general-purpose technologies reshape labour markets and firm dynamics (Felten et al., 2021, Fossen et al., 2024). Most of this debate focuses on *workers*: which jobs are at risk, and for whom. But a parallel question may be equally consequential: does AI alter the incentives to *start* a business?

Fossen et al. (2024) examine AI adoption and self-employment at the individual level and find positive effects concentrated among high-skilled workers. Chatterji et al. (2025) provide a conceptual treatment of how transformative AI reshapes firm organisation, adoption incentives, and competitive dynamics—a natural complement to the reduced-form evidence presented here. Felten et al. (2021) construct the AIOE/AIIE index by combining O*NET task measures with AI capability scores. Haltiwanger et al. (2020) describe the Census Bureau BFS and demonstrate its value for studying business dynamism and entry margins. Brynjolfsson et al. (2018) provide a conceptual framework for mapping task-level AI capabilities to occupational and industry outcomes.

Lower entry costs, improved productivity for high-skilled tasks, and easier access to specialised knowledge may all reduce barriers to entrepreneurship in AI-exposed sectors (Brynjolfsson et al., 2018). Conversely, if AI primarily benefits large incumbents, competitive pressures could *raise* barriers to entry. Distinguishing between these channels requires variation both across sectors and over time—variation the public release of ChatGPT conveniently provides.

2 Data

2.1 Business Formation Statistics

The Census Bureau BFS provides monthly, seasonally adjusted counts of new Employer Identification Number (EIN) applications at the 2-digit NAICS level, derived from IRS Form SS-4 filings (Haltiwanger et al., 2020). EIN applications are a leading indicator of actual busi-

ness formation: roughly half of high-propensity applicants open within eight quarters of filing. We use the official seasonally adjusted series directly and index each sector to January 2020 = 100, yielding a panel observed from January 2006 through early 2026. The BFS publishes 20 sector-level series (excluding the unclassified "No NAICS" category): Manufacturing (NAICS 31–33), Retail Trade (44–45), and Transportation & Warehousing (48–49) are each published as single combined codes rather than individual 2-digit breakdowns. Public Administration (NAICS 92) is present in the BFS but has no corresponding AIOE score and is therefore excluded from the regression sample, leaving 19 sectors. The treatment date is November 2022, the public launch of ChatGPT.

2.2 AI Exposure

Following Felten et al. (2021), we use the AI Industry Exposure (AIIE) index, which maps occupational AI exposure scores to 4-digit NAICS industries through BLS employment weights and aggregates to 2-digit NAICS by employment-weighted averaging. The index is constructed from 10 AI application domains benchmarked against O*NET task descriptions and captures the degree to which a sector's occupational mix is susceptible to AI automation or augmentation. Finance and Insurance (NAICS 52, AIIE = 2.05), Professional Services (NAICS 54, AIIE = 1.59), and Information (NAICS 51, AIIE = 1.27) rank highest; Agriculture (NAICS 11, AIIE = −1.56), Transportation (NAICS 49, AIIE = −1.31), and Construction (NAICS 23, AIIE = −1.00) rank lowest. This cross-sectional variation—which is fixed before the ChatGPT shock—is the key source of identification. Table 1 reports the AIIE score alongside summary statistics of the BFS series for each sector in the sample.

Table 1: AI Industry Exposure and Business Formation Statistics by Sector

Sector	NAICS	AIIE	Mean	SD	Min	Max	N
			Monthly BA (thousands)				(months)
Finance & Insurance	52	2.05	14.0	3.6	8.4	22.0	261
Management of Companies	55	1.70	2.2	1.3	0.9	6.6	261
Professional, Scientific, Technical Svcs	54	1.59	38.2	11.7	19.8	76.7	261
Educational Services	61	1.32	4.6	1.5	2.4	8.6	261
Information	51	1.27	6.1	1.8	3.0	11.4	261
Health Care & Social Assistance	62	0.56	18.3	5.6	10.2	33.1	261
Real Estate & Rental	53	0.51	16.9	5.4	8.9	26.5	261
Wholesale Trade	42	0.26	8.9	1.3	6.1	16.2	261
Utilities	22	0.13	0.3	0.2	0.1	0.8	261
Arts, Entertainment, & Recreation	71	0.10	9.5	2.2	5.3	15.4	261
Retail Trade (NAICS 44–45)	RET	-0.01	45.2	24.7	20.1	131.0	261
Other Services	81	-0.03	26.6	10.0	12.7	62.8	261
Admin. & Support Services	56	-0.21	20.3	7.2	10.9	41.9	261
Manufacturing (NAICS 31–33)	MNF	-0.57	5.7	1.3	3.7	10.5	261
Mining, Quarrying, Oil & Gas	21	-0.68	0.6	0.1	0.3	1.4	261
Transportation & Warehousing (48–49)	TW	-0.77	18.8	9.7	7.9	44.0	261
Accommodation & Food Services	72	-0.93	15.6	5.5	8.5	31.6	261
Construction	23	-1.00	29.7	9.1	16.0	52.9	261
Agriculture, Forestry, Fishing	11	-1.56	3.5	1.5	1.9	19.4	261

Notes: AIIE = AI Industry Exposure index from Felten et al. (2021), constructed from O*NET task descriptions and aggregated from 4-digit to 2-digit NAICS by employment-weighted averaging. Mean, SD, Min, and Max refer to monthly seasonally adjusted business applications (thousands) over the full sample period (January 2006 to early 2026). N is the number of monthly observations. Sectors sorted by AIIE score descending.

3 Empirical Strategy

[Empirical strategy placeholder.]

Our baseline specification is equation (1). The identifying assumption is that, absent the ChatGPT launch, business applications would have evolved in parallel across sectors regardless of their AI exposure. We probe this assumption with an event-study specification that replaces Post_t with a full set of month-relative-to-November-2022 indicators interacted with AI Exposure_i. Parallel pre-trends would be consistent with the identification assumption.

Standard errors are clustered at the 2-digit NAICS sector level using `fixest` (Bergé, 2018).

We probe robustness along two dimensions. First, we substitute the outcome variable with the full suite of BFS series, which span the entire formation funnel. On the applications side: corrected business applications (CBA), which purge filings unlikely to yield an employer business; high-propensity applications (HBA); and applications with planned wages (WBA). On the formations side: businesses actually formed within four quarters (BF4Q) and eight quarters (BF8Q) of filing; Census-projected formations at both horizons (PBF4Q, PBF8Q); and seasonally-adjusted formation counts (SBF4Q, SBF8Q). Finally, the median number of weeks from application to formation (DUR4Q, DUR8Q) provides a complementary timing dimension: a negative coefficient would indicate that AI exposure compresses the application-to-opening lag, consistent with lower operational setup costs. If AI lowers barriers to entry, the effect should be visible across all stages of the formation funnel. Second, we re-estimate the model at the state level, replacing sector fixed effects with state fixed effects. The outcome is the log of total seasonally adjusted business applications aggregated across all sectors within a state, and the AI exposure measure is the AI Geographic Exposure (AIGE) index from Felten et al. (2021) Appendix C, which is constructed directly from the occupational mix of each state and its workers' AI exposure. This specification absorbs permanent state-level differences in entrepreneurial activity and exploits cross-state variation in AI exposure that is independent of the sector composition used in the main analysis.

Table 2: Cross-Sectional Regression: AI Exposure and Change in Business Applications

	OLS	WLS
AI Exposure (AIOE, normalised)	54.175***	41.815***
	(15.637)	(12.767)
Constant	-8.852	-3.093
	(8.729)	(7.047)
Sectors	19	19
R^2	0.414	0.387

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
OLS and WLS (weighted by sector mean BA, Nov 2022) estimates. Outcome: percentage change in business applications from November 2022 to the most recent month available. Each observation is a 2-digit NAICS sector ($N = 19$).

4 Results

4.1 Descriptive Evidence

4.2 Difference-in-Differences

Figure 2 plots the percentage change in business applications from November 2022 to the most recent month available against each sector's AI exposure.

Table 2 formalises the scatter with a cross-sectional regression of the percentage change on normalised AIOE. Column (1) is unweighted OLS; column (2) weights observations by pre-period mean business applications, giving larger sectors proportionally more influence. The slope is positive and statistically significant in both specifications, and the weighted estimate is somewhat larger, indicating that the positive relationship between AI exposure and application growth is stronger among economically larger sectors.

Table 3 reports the DiD estimates. All three columns pool the entire post-November 2022 period into a single *Post* indicator, with columns (1)–(3) using log business applications, the indexed BA outcome, and raw BA levels respectively.

To translate the coefficient into economically interpretable magnitudes, Table 4 reports

Table 3: Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) BA Level
Post × AI Exposure	0.209	30.151	-1607.456
	(0.227)	(30.712)	(7157.107)
Observations	4959	4959	4959
R^2	0.982	0.690	0.834
Sector FE	✓	✓	✓
Month FE	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by 2-digit NAICS sector.

the implied percentage difference in business applications relative to a sector at the 25th percentile of AI exposure, evaluated at selected percentiles of the AIOE distribution. A sector at the median of AI exposure is estimated to have had [X]% more applications than a P25 sector in the post-ChatGPT period; the gap widens to [X]% at the 75th percentile and [X]% at the 90th.

Table 4: Implied Effect of AI Exposure on Business Applications

Percentile	AIOE (normalised)	Effect vs. P25 sector (%)
P10	0.166	-1.6%
P25	0.244	+0.0%
P50	0.424	+3.8%
P75	0.580	+7.0%
P90	0.828	+12.2%

Notes: Implied percentage difference in business applications in the post-ChatGPT period relative to a sector at the 25th percentile of normalised AI Industry Exposure (AIOE). Computed as $(\text{AIOE}_p - \text{AIOE}_{P25}) \times \hat{\beta} \times 100$, where $\hat{\beta}$ is the DiD estimate from column (1) of Table 3.

4.3 Event Study

Figure 3 presents the event-study coefficients. The pre-period estimates ($t < 0$) are small and statistically indistinguishable from zero, supporting the parallel-trends assumption. Post-launch, coefficients rise gradually before the 95 % confidence interval first clears zero at $t = +18$ (May 2024, approximately 18 months after the ChatGPT launch) and remains above zero through the end of the sample, indicating a persistent differential effect of AI exposure on business formation. This pattern is consistent with a diffusion mechanism: entrepreneurs require time to learn the tools, identify viable business models, and translate that into actual EIN filings. A visible second inflection around $t = +30$ (May 2025) is consistent with further acceleration as agentic AI tools became more widely available, though the gradual nature of that development makes a clean identification of a second event date difficult.

Figure 3 reveals a more nuanced two-phase pattern. A handful of coefficients are statistically significant in the first months immediately following the ChatGPT release ($t = 0$, $+1$, $+3$, $+4$, $+5$), suggesting an early, tentative response. These effects then recede into insignificance across roughly a year ($t = +6$ through $t = +16$), before re-emerging persistently from $t = +17$ (April 2024, $p < 0.10$) and $t = +18$ (May 2024, $p < 0.05$) onward. From May 2024, every subsequent monthly coefficient is statistically significant at the 5 % level or better, and point estimates grow monotonically through the end of the sample, reaching 0.48 at $t = +36$. The gap between the initial signal and the onset of persistent significance—roughly 12 months—is consistent with a learning-and-diffusion account: the early uptick reflects awareness and experimentation by first-movers, while the later, durable response reflects a broader population of entrepreneurs who required time to identify viable AI-enabled business models and convert that into EIN filings.

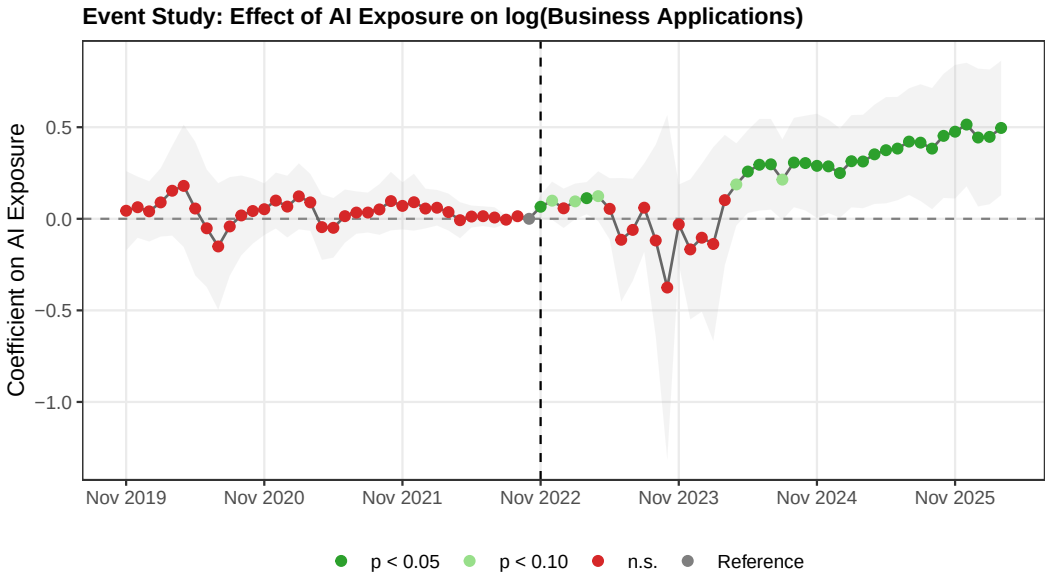

Figure 3: Event-study estimates: effect of AI exposure on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on sector-clustered standard errors.

4.4 Robustness Checks

Alternative BFS series

Table 5 replicates the baseline DiD specification (column (1) of Table 3) using three alternative BFS outcome variables. Column (2) replaces total business applications with high-propensity applications (HBA), defined by the Census Bureau as those most likely to become active employer businesses. Column (3) uses applications with planned wages (WBA), which signal explicit intent to hire at the time of EIN filing. Column (4) uses the count of businesses that actually formed within four quarters of application (BF4Q), the realised formation margin. A consistent pattern across the formation funnel—from intent (WBA) to high-quality applications (HBA) to actual formations (BF4Q)—strengthens the interpretation that AI exposure shifted both the quantity and quality of entrepreneurial activity, not merely the propensity to file. We additionally examine formations within eight quarters (BF8Q) in the event-study figures; because the Census requires a full eight-quarter follow-up window

before publishing BF8Q, the series is right-truncated approximately two years before the data vintage and does not have sufficient post-period coverage for a clean DiD estimate. The event-study coefficients are nevertheless informative: comparing the BF4Q and BF8Q panels reveals whether the additional entry in high-AI-exposure sectors converts to operating businesses quickly or gradually.

Figure 4 presents the corresponding event-study coefficients for all four series. The pre-period estimates are small and insignificant across every panel, ruling out differential pre-trends. Post-launch dynamics are qualitatively consistent: coefficients drift upward at similar timing and reach statistical significance in the same window regardless of which formation margin is used as the outcome. The BF4Q panel is shifted rightward relative to applications panels, reflecting the mechanical lag between filing and actual business opening, and provides perhaps the cleanest evidence that the effect translates into realised entry rather than merely intent.

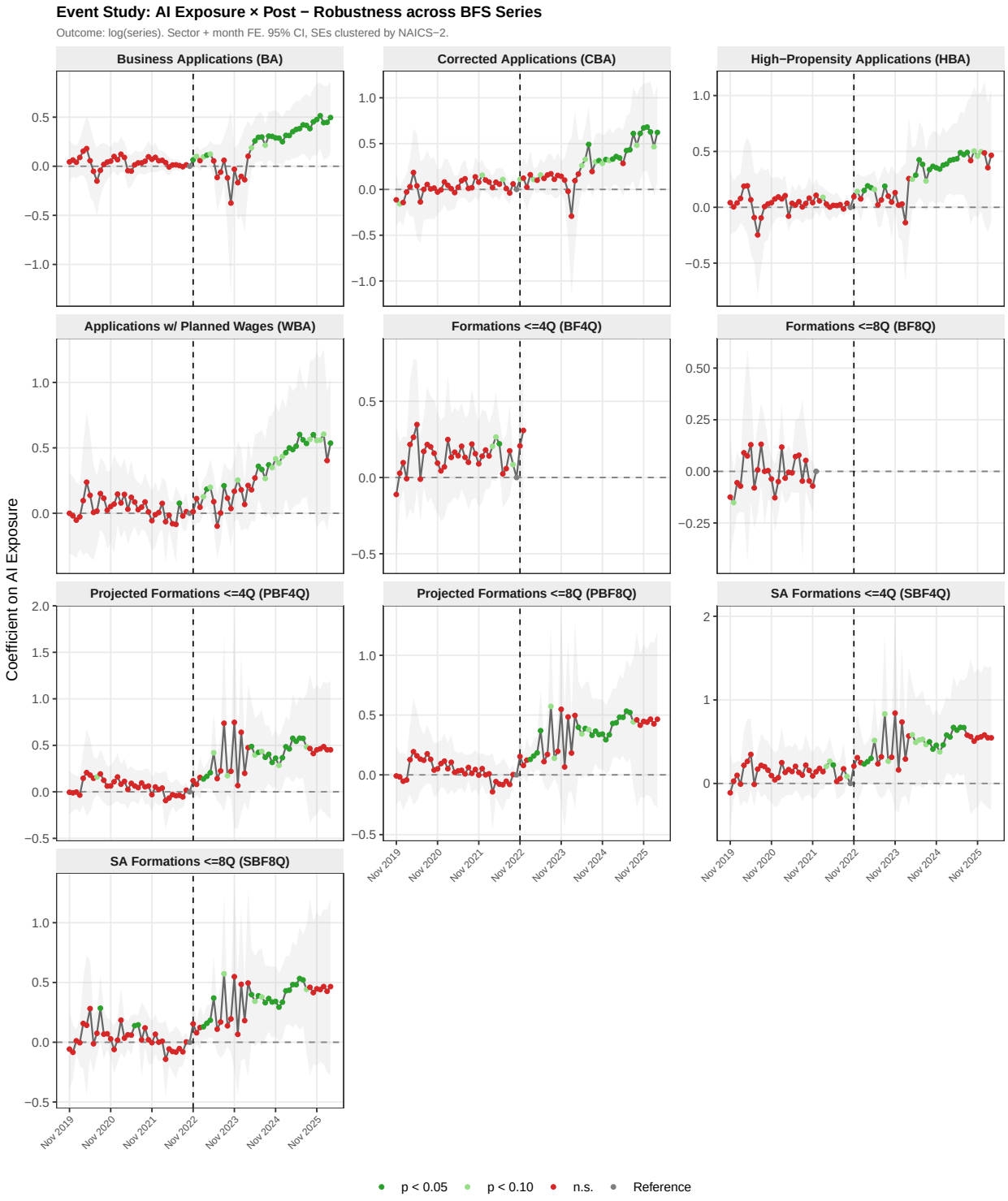

Figure 4: Event-study estimates for four BFS series. Each panel plots the interaction coefficient between AI exposure and a month-relative-to-November-2022 indicator. BA = total applications; CBA = corrected applications; HBA = high-propensity applications; WBA = applications with planned wages; BF4Q/BF8Q = formations within 4/8 quarters; PBF4Q/PBF8Q = projected formations; SBF4Q/SBF8Q = seasonally-adjusted formations; DUR4Q/DUR8Q = median weeks to formation (levels, not logs). Shaded bands are 95 % confidence intervals based on sector-clustered standard errors. All specifications include sector and month fixed effects. Dashed vertical line marks November 2022 ($t = 0$).

Table 5: Robustness: Alternative BFS Series

	(1) log(BA)	(2) log(CBA)	(3) log(HBA)	(4) log(WBA)	(5) log(BF4Q)
Post × AI Exposure	0.209	0.426	0.212	0.287	0.377
	(0.227)	(0.292)	(0.291)	(0.267)	(0.283)
Observations	4959	4959	4959	4959	4216
R^2	0.982	0.980	0.982	0.984	0.988
Sector FE	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01

Standard errors clustered by 2-digit NAICS sector. All specifications include sector and month fixed effects. BA = total business applications; HBA = high-propensity business applications; WBA = applications with planned wages; BF4Q = businesses formed within four quarters of application.

Appendix **??** reports a state-level robustness check using the AI Geographic Exposure (AIGE) index of Felten et al. (2021) and state $\times$ month panels. The state-level estimates are statistically insignificant, likely because state aggregation dilutes the within-sector AI variation that drives the main results; the sector-level estimates remain the primary evidence.

5 Conclusion

[Conclusion placeholder.]

A State-Level Robustness

Table 6 presents the state-level DiD estimates. The unit of observation is a state-month cell, with the outcome being log total seasonally adjusted business applications aggregated across all sectors within the state. The AI exposure measure is the AIGE index of Felten et al. (2021), which captures the AI exposure of a state's workforce through its occupational mix. State fixed effects absorb permanent cross-state differences in entrepreneurial culture, regulatory environment, and industrial composition; month fixed effects absorb national macro shocks as before. Standard errors are clustered at the state level (56 clusters).

Figure 5 provides raw descriptive evidence at the state level, analogous to Figure **??** for sectors. States are sorted into quartiles by their AIGE index and the cross-state mean of indexed business applications (January $2020 = 100$) is plotted for each quartile.

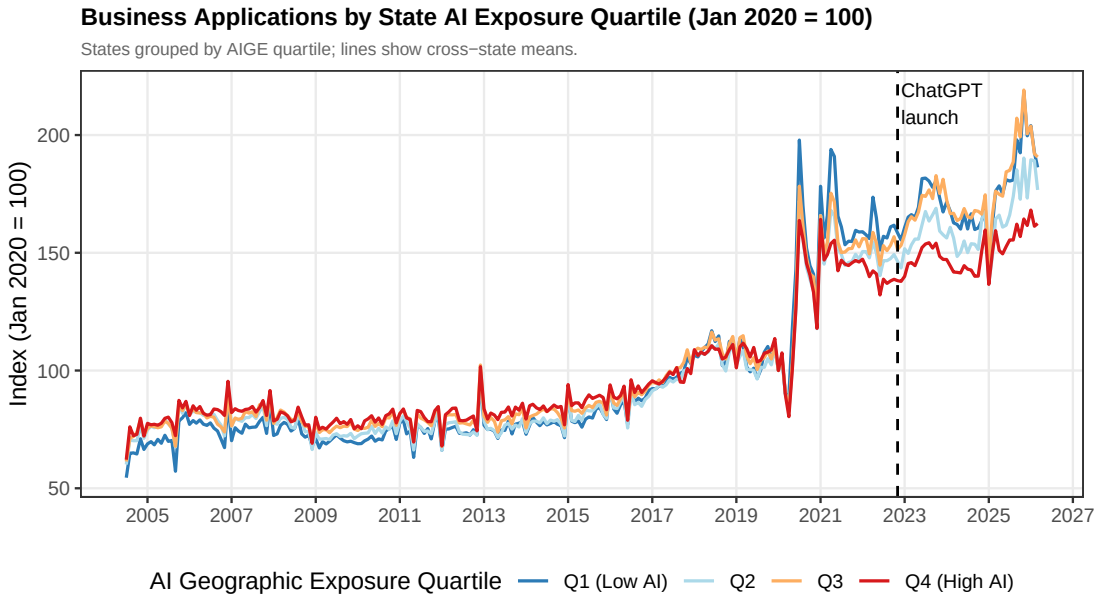

Figure 5: Mean business applications (index, January 2020 = 100) by AI Geographic Exposure (AIGE) quartile across US states and territories. Q1 denotes the lowest-exposure quartile; Q4 the highest. Lines show simple cross-state means within each quartile. Dashed vertical line marks November 2022 (ChatGPT launch).

Column (1) shows the single-period estimate; column (2) uses the indexed BA outcome; column (3) splits the post period as in the main analysis. The state-level estimates are statistically insignificant across all specifications, in contrast to the sector-level results in Table 3.

Figure 6 shows the corresponding event-study coefficients, and Figure 7 repeats the exercise across the four BFS series published at the state level (BA, HBA, WBA, CBA; formation series are national-only). Pre-period coefficients are flat across all panels, consistent with parallel pre-trends, but the post-launch coefficients are imprecise and do not reach conventional significance levels in any series.

The absence of a detectable state-level effect likely reflects two compounding limitations. First, states are highly diversified: even a state with a high-AIGE workforce will contain many low-AI sectors that dilute the aggregate application signal. The sector-level design exploits within-sector variation in AI exposure that is averaged away when outcomes are aggregated to the state level. Second, the cross-state dispersion of AIGE is considerably

Table 6: State-Level Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) Two-period
$\text{Post}_{\text{ChatGPT}} \times$ AI Exposure	-0.445	-67.191	-0.423
	(0.274)	(42.331)	(0.265)
Observations	13572	13572	13572
R^2	0.986	0.825	0.986
State FE	✓	✓	✓
Month FE	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by state. AI exposure is the AI Geographic Exposure (AIGE) index from Felten et al. (2021) Appendix C, constructed from occupation-level AIOE scores weighted by the state employment-mix across occupations. $\text{Post}_{\text{ChatGPT}}$: Nov 2022 – Apr 2025. $\text{Post}_{\text{Agentic}}$: May 2025 onwards.

smaller than the cross-sector dispersion of AIOE, reducing statistical power. Together, these features make the state-level panel a demanding test; the sector-level estimates remain the primary evidence.

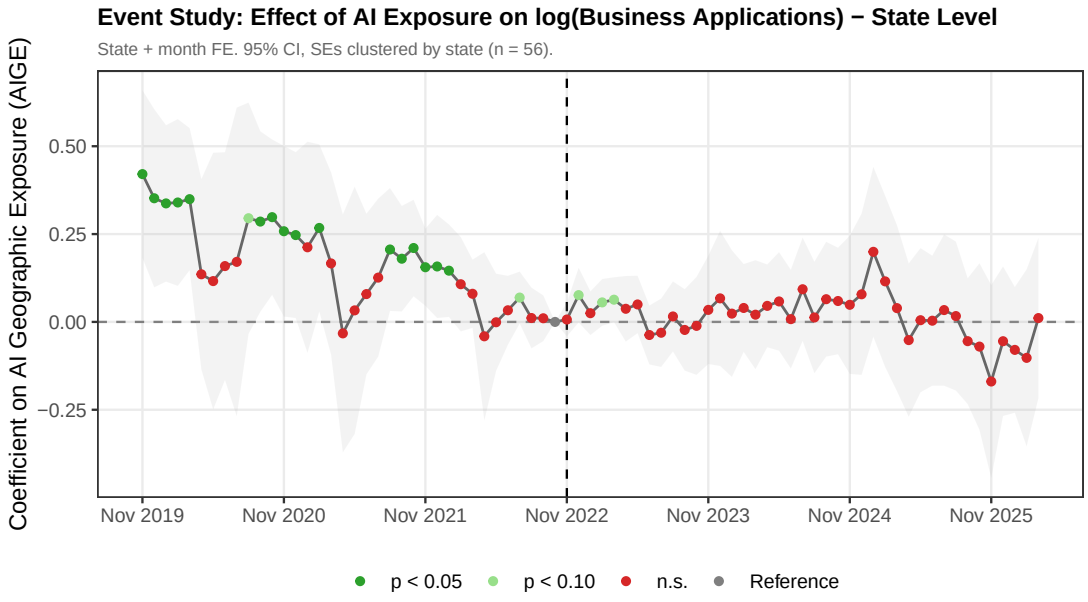

Figure 6: State-level event-study estimates: effect of AI Geographic Exposure (AIGE) on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on state-clustered standard errors (56 clusters). State and month fixed effects included. Coefficients are statistically insignificant throughout.

Figure 7 replicates the state-level event study across the four BFS series available at state level.

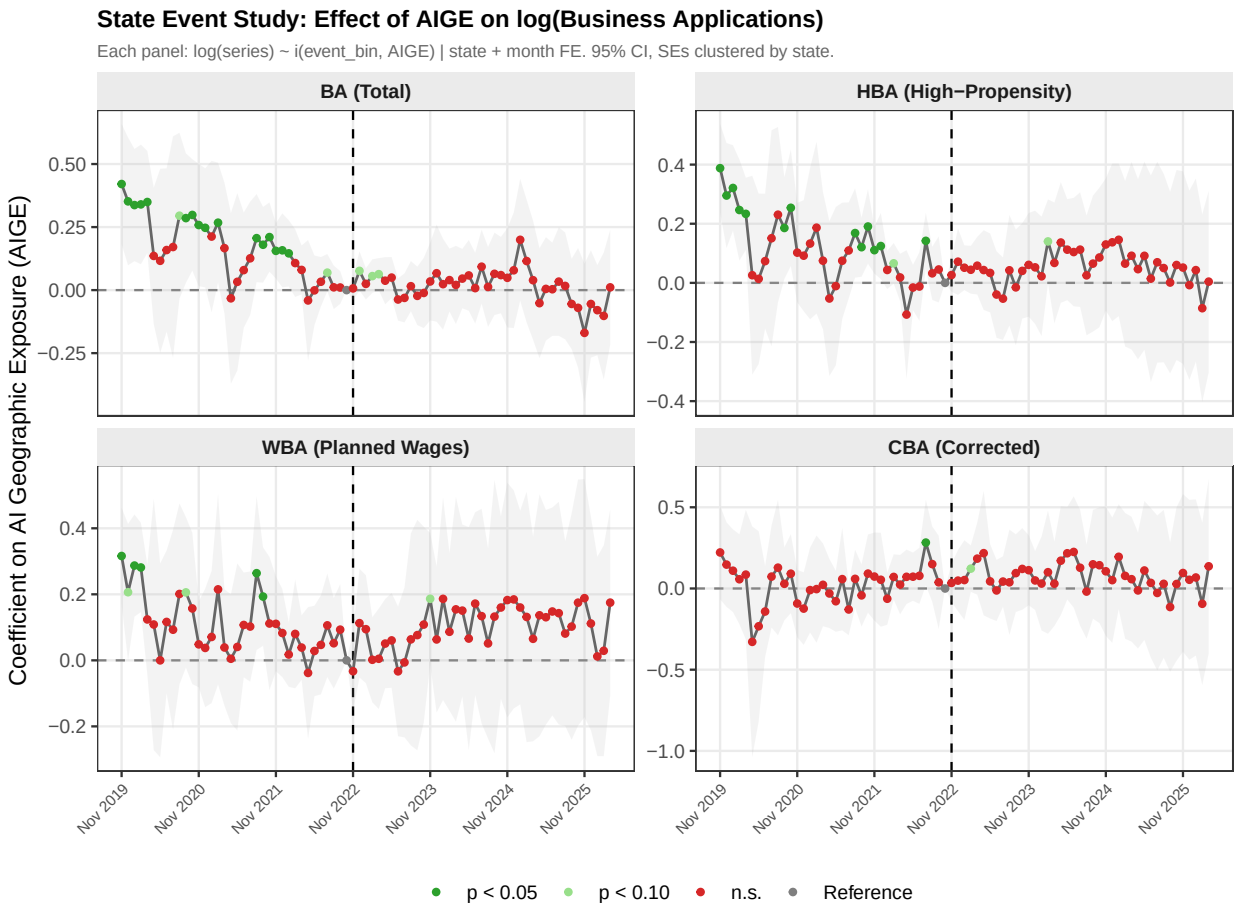

Figure 7: State-level event-study estimates for four BFS series. Each panel plots the interaction coefficient between AI Geographic Exposure (AIGE) and a month-relative-to-November-2022 indicator, estimated on the state × month panel. BA = total applications; HBA = high-propensity applications; WBA = applications with planned wages; CBA = corrected applications. Business formation series (BF4Q, BF8Q, etc.) are not published at state level by the Census Bureau. Shaded bands are 95 % confidence intervals based on state-clustered standard errors (56 clusters). All specifications include state and month fixed effects. Dashed vertical line marks November 2022 ($t = 0$).

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